

CHAPTER 4: ANIMAL KINGDOM

Complete Detailed Academic Notes • Diagrams • Flowcharts • Solved Exercises

100%

Exam-Oriented Coverage

4.1 Basis of Classification

- Levels of Organisation
- Symmetry
- Germinal Layers
- Coelom & Segmentation
- Notochord

4.2 Non-Chordates

- Porifera to Aschelminthes
- Annelida & Arthropoda
- Mollusca & Echinodermata
- Hemichordata

4.3 Chordates & Vertebrata

- Chordata vs Non-Chordata
- Protochordates
- Cyclostomata & Chondrichthyes
- Osteichthyes, Amphibia, Reptilia
- Aves & Mammalia

4.4 Revision & Exercises

- Comprehensive Phylum Table
- Chapter Summary
- Solved NCERT Exercises (1-15)
- Mnemonics & Key Highlights

CHAPTER MIND MAP — KINGDOM ANIMALIA CLASSIFICATION HIERARCHY

KINGDOM ANIMALIA (Multicellular Heterotrophs)

Cellular Level

(Mostly Asymmetrical, Acoelomate)

Phylum Porifera

Tissue / Organ / Organ-System Level

Radial Symmetry

(Acoelomate)

Coelenterata (Cnidaria)

Ctenophora

Bilateral Symmetry

Acoelomate
Platyhelminthes**Pseudocoelomate**
Aschelminthes**True Coelomate**
Annelida, Arthropoda, Mollusca,
Echinodermata*, Hemichordata,
Chordata

* Note: Adult Echinoderms show Radial symmetry, while larvae exhibit Bilateral symmetry.

In spite of vast differences in form and structure among over a million animal species, fundamental features such as cell arrangement, body symmetry, nature of coelom, embryonic layers, and digestive/circulatory patterns form the basis of classification.

4.1.1 Levels of Body Organisation

- 1. Cellular Level:** Loose cell aggregates with basic division of labour. **Ex: Sponges (Porifera)** .
- 2. Tissue Level:** Cells performing same function organized into distinct tissues. **Ex: Coelenterates, Ctenophores** .
- 3. Organ Level:** Tissues grouped into specialized organs. **Ex: Platyhelminthes** .
- 4. Organ-System Level:** Organs associated to form functional physiological systems. **Ex: Annelids, Arthropods, Molluscs, Echinoderms, Chordates** .

Digestive System Complexity:

- *Incomplete*: Single opening for mouth & anus (e.g., Platyhelminthes).
- *Complete*: Two distinct openings (mouth and anus).

Circulatory System Types:

- *Open Type*: Blood pumped into tissue spaces/sinuses (cells bathed directly).
- *Closed Type*: Blood circulated continuously inside closed vessels.

4.1.2 Body Symmetry Patterns

Asymmetrical: No plane passing through the center can divide the body into equal halves.

Examples: Most Sponges.

Radial Symmetry: Any plane passing through central axis divides organism into identical halves.

Examples: Coelenterates, Ctenophores, Echinoderms (Adults).

Bilateral Symmetry: Body divided into identical left and right halves in only ONE single plane.

Examples: Annelids, Arthropods, Molluscs, Chordates.

4.1.3 Embryonic Germ Layers

Diploblastic Animals: Cells arranged in 2 embryonic layers — outer **Ectoderm** and inner **Endoderm**, with undifferentiated non-cellular **Mesoglea** in between.

Ex: Coelenterates, Ctenophores.

Triploblastic Animals: Developing embryo possesses a 3rd cellular layer, **Mesoderm**, between ectoderm & endoderm.

Ex: Platyhelminthes to Chordates.

4.1.4 Nature of Body Cavity (Coelom)

Acoelomate

Body cavity absent; mesoderm fills space.

Platyhelminthes

Pseudocoelomate

Mesoderm present as scattered pouches between ecto & endoderm.

Aschelminthes

Coelomate (Eucoelomate)

True coelom lined completely by mesoderm on both sides.

Annelida to Chordata

4.1.5 Metameric Segmentation: External & internal division into serial segments with repetition of organs (e.g., Earthworm metamerism).

4.1.6 Notochord: Mesodermally derived solid rod-like structure formed on dorsal side during embryonic development (Chordates vs Non-chordates).

4.2.1 Phylum Porifera

Sponges

- **Habitat/Symmetry:** Primitive marine, mostly asymmetrical, cellular level.
- **Water Canal System:** Water enters via **Ostia** into **Spongocoel** cavity and exits through **Osculum**.
- **Functions:** Food gathering, gas exchange, waste removal.
- **Choanocytes (Collar Cells):** Line spongocoel & canals.
- **Digestion:** Intracellular. **Skeleton:** Spicules or spongin.
- **Reproduction:** Hermaphrodite; Asexual (fragmentation), Sexual (gametes). Internal fertilisation; Indirect development.

Examples: *Sycon* (Scypha), *Spongilla* (Fresh water sponge), *Euspongia* (Bath sponge).

4.2.2 Phylum Cnidaria

Coelenterates

- **Habitat:** Aquatic, mostly marine, sessile/free-swimming, radial.
- **Cnidoblasts/Cnidocytes:** Stinging cells with nematocysts on tentacles for anchorage, defense, prey capture.
- **Structure:** Tissue level, diploblastic, gastro-vascular cavity with single mouth on hypostome. CaCO_3 skeleton in corals.
- **Dimorphism:**
 1. *Polyp*: Sessile, cylindrical (e.g., *Hydra*, *Adamsia*).
 2. *Medusa*: Umbrella-shaped, free-swimming (e.g., *Aurelia*).
- **Metagenesis (Alternation of Gen):** Polyps produce medusae asexually; Medusae produce polyps sexually (e.g., *Obelia*).

Examples: *Physalia* (Portuguese man-of-war), *Adamsia* (Sea anemone), *Pennatula* (Sea-pen), *Gorgonia* (Sea-fan), *Meandrina* (Brain coral).

4.2.3 Phylum Ctenophora

Comb Jellies

- **General:** Exclusively marine, radially symmetrical, diploblastic, tissue level.
- **Comb Plates:** 8 external rows of ciliated comb plates used for locomotion.
- **Bioluminescence:** Ability to emit light is well-marked.
- **Digestion:** Both extracellular and intracellular.
- **Reproduction:** Bisexual (sexes not separate); ONLY sexual reproduction; External fertilisation; Indirect development.

Examples: *Pleurobrachia*, *Ctenoplana*.

4.2.4 Platyhelminthes

Flatworms

- **Form:** Dorso-ventrally flattened body. Bilateral, triploblastic, acoelomate, organ level.
- **Parasitic Adaptations:** Endoparasites, hooks and suckers present, direct body surface absorption.
- **Flame Cells:** Specialised cells for osmoregulation and excretion.
- **Reproduction:** Monoecious; Internal fertilisation; Many larval stages. High regeneration in *Planaria*.

Examples: *Taenia* (Tapeworm), *Fasciola* (Liver fluke).

4.2.5 Aschelminthes

Roundworms

- **Form:** Circular in cross-section. Organ-system level, bilateral, triploblastic, **Pseudocoelomate**.
- **Digestive Tract:** Complete alimentary canal with well-developed **muscular pharynx**.
- **Excretion:** Excretory tube removes wastes via excretory pore.
- **Sexual Dimorphism:** Dioecious (separate sexes); Females often longer than males. Internal fertilisation.

Examples: *Ascaris* (Roundworm), *Wuchereria* (Filaria worm), *Ancylostoma* (Hookworm).

4.2.6 Phylum Annelida

Segmented Worms

- **Segmentation:** Body metamerically segmented into metameres (ring-like segments).
- **Locomotion:** Longitudinal & circular muscles. Lateral appendages **Parapodia** in aquatic forms (*Nereis*) for swimming.
- **Systems:** Closed circulatory system. **Nephridia** for osmoregulation & excretion. Double ventral nerve cord with paired ganglia.
- **Sexuality:** *Nereis* is dioecious; Earthworms & Leeches are monoecious.

Examples: *Nereis*, *Pheretima* (Earthworm), *Hirudinaria* (Blood sucking leech).

4.2.7 Phylum Arthropoda

Largest Phylum (>2/3 of species)

- **Body Regions:** Head, Thorax, and Abdomen covered by a tough **Chitinous Exoskeleton**.
- **Appendages:** Jointed appendages (*arthros* = joint, *poda* = appendage).
- **Respiration:** Gills, Book Gills, Book Lungs, or Tracheal System.
- **Circulation & Excretion:** Open circulatory system; Excretion via **Malpighian Tubules**.
- **Sensory Organs:** Antennae, simple & compound eyes, **Statocysts** (balancing organs).
- **Reproduction:** Mostly dioecious, usually internal fertilisation, oviparous.

Economic: *Apis* (Honey bee), *Bombyx* (Silkworm), *Laccifer* (Lac insect)

Vectors: *Anopheles*, *Culex*, *Aedes* mosquitoes

Pest: *Locusta* (Locust)

Living Fossil: *Limulus* (King crab)

4.2.8 Phylum Mollusca

Second Largest Phylum

- **Structure:** Soft, unsegmented body covered by a hard **Calcereous Shell**. Divided into **Head**, **Muscular Foot**, and **Visceral Hump**.
- **Mantle & Mantle Cavity:** Spongy layer of skin forms a mantle over visceral hump. Space between hump and mantle contains **Feather-like Gills** (respiratory & excretory).
- **Radula:** File-like rasping organ in mouth for feeding.
- **Reproduction:** Dioecious, oviparous, indirect development.

Examples: *Pila* (Apple snail), *Pinctada* (Pearl oyster), *Sepia* (Cuttlefish), *Loligo* (Squid), *Octopus* (Devil fish), *Aplysia* (Sea-hare), *Dentalium* (Tusk shell), *Chaetopleura* (Chiton).

4.2.9 Phylum Echinodermata

Spiny Bodied

- **Skeleton & Habitat:** Endoskeleton of calcareous ossicles. All marine.
- **Symmetry Key Point:** Adult echinoderms are **Radially symmetrical**, but larvae are **Bilaterally symmetrical**.
- **Water Vascular System:** Unique system for locomotion, capture/transport of food, and respiration.
- **Excretory System:** ABSENT.
- **Reproduction:** Dioecious, external fertilisation, indirect development with free-swimming larva.

Examples: *Asterias* (Star fish), *Echinus* (Sea urchin), *Antedon* (Sea lily), *Cucumaria* (Sea cucumber), *Ophiura* (Brittle star).

4.2.10 Phylum Hemichordata

Worm-like Marine

- **Taxonomic Note:** Previously considered a sub-phylum of Chordata, now placed as a separate Non-chordate phylum.
- **Stomochord:** Rudimentary structure in collar region similar to notochord.
- **Body Regions:** Cylindrical body composed of anterior **Proboscis**, **Collar**, and long **Trunk**.
- **Organs:** Open circulation, respiration via gills, excretion via **Proboscis Gland**.
- **Reproduction:** Dioecious, external fertilisation, indirect development.

Examples: *Balanoglossus*, *Saccoglossus*.

4 Fundamental Chordate Features

1. Presence of **Notochord**.
2. Dorsal, hollow, single **Nerve Cord**.
3. Paired **Pharyngeal Gill Slits**.
4. Post-anal tail & Closed circulatory system.

#	Feature	Chordates	Non-Chordates
1	Notochord	Present	Absent
2	Central Nervous System	Dorsal, hollow, single	Ventral, solid, double
3	Pharyngeal Gill Slits	Present (perforated)	Absent
4	Heart Position	Ventral	Dorsal (if present)
5	Post-Anal Tail	Present	Absent

Protochordata & Cyclostomes

- **Urochordata:** Notochord in larval tail only (e.g., *Ascidia*, *Salpa*, *Doliolum*).
- **Cephalochordata:** Notochord extends head to tail throughout life (e.g., *Branchiostoma* / *Amphioxus*).

Class Cyclostomata:

Jawless ectoparasites on fish; 6-15 pairs gill slits; sucking circular mouth; cartilaginous cranium. Anadromous migration (spawn in fresh water, die; larvae return to ocean).

Ex: *Petromyzon* (Lamprey), *Myxine* (Hagfish).

Super Class Pisces

1. Chondrichthyes (Cartilaginous): Ventral mouth; persistent notochord; no operculum; placoid scales; air bladder ABSENT (constant swimming); claspers in males; viviparous.

Ex: *Scoliodon* (Dog fish), *Pristis* (Saw fish), *Torpedo* (Electric ray), *Trygon* (Poison sting ray).

2. Osteichthyes (Bony): Terminal mouth; 4 pairs gills with operculum; cycloid scales; air bladder PRESENT (buoyancy); external fertilisation; oviparous.

Ex: *Exocoetus*, *Hippocampus*, *Labeo*, *Catla*, *Betta*.

Amphibia & Reptilia

Class Amphibia: Dual life; moist skin without scales; 3-chambered heart (2A + 1V); cloaca present; cold-blooded (poikilotherms).

Ex: *Bufo*, *Rana*, *Hyla*, *Salamandra*, *Ichthyophis* (limbless).

Class Reptilia: Creeping/crawling; dry cornified skin with epidermal scales/scutes; 3-chambered heart (4 in Crocodiles); poikilotherms; internal fertilisation.

Ex: *Chelone*, *Chameleon*, *Crocodilus*, *Naja*, *Vipera*.

Aves & Mammalia

Class Aves (Birds): Feathers, forelimbs modified to wings; pneumatic (hollow) bones; crop & gizzard; 4-chambered heart; warm-blooded (homoiothermous).
Ex: *Corvus*, *Columba*, *Psittacula*, *Struthio*, *Pavo*.

Class Mammalia: Mammary glands (milk producing); body hair; external ears (pinnae); heterodont teeth; 4-chambered heart; homoiothermous; viviparous (except Platypus).

Ex: *Ornithorhynchus* (oviparous Platypus), *Macropus*, *Pteropus*, *Panthera*.

TABLE 4.2 COMPREHENSIVE PHYLUM COMPARISON & SOLVED NCERT EXERCISES

NCERT Pages 51–54

Phylum	Level of Org.	Symmetry	Coelom	Segm.	Digestive	Circulatory	Respiratory	Distinctive Features
Porifera	Cellular	Various	Absent	Absent	Absent	Absent	Absent	Body with pores (ostia) and canal system; Choanocytes.
Cnidaria	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Cnidoblasts/Cnidocytes present; Polyp & Medusa forms.
Ctenophora	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	8 ciliated comb plates for locomotion; Bioluminescence.
Platyhelminthes	Organ & Org-sys	Bilateral	Absent	Absent	Incomplete	Absent	Absent	Flat body; Hooks and suckers; Flame cells for excretion.
Aschelminthes	Organ-system	Bilateral	Pseudocoelom	Absent	Complete	Absent	Absent	Roundworm shape; Muscular pharynx; Sexual dimorphism.
Annelida	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Absent	Metameric segments; Parapodia in Nereis; Nephridia.
Arthropoda	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Exoskeleton of chitin; Jointed appendages; Malpighian tubules.
Mollusca	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	Calcareous shell; Visceral hump & mantle; Radula.
Echinodermata	Organ-system	Radial (Adult)	Coelomate	Absent	Complete	Present	Present	Spiny skin; Water vascular system; Bilateral larvae.
Hemichordata	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	Worm-like with Proboscis, Collar & Trunk; Stomochord.
Chordata	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Notochord; Dorsal hollow nerve cord; Pharyngeal gill slits.

★ SOLVED NCERT EXERCISE HIGHLIGHTS (Q1 – Q15)

Q8. Water vascular system is characteristic of:Ans: **(c) Echinodermata.** (Helps in locomotion, food capture, and respiration).**Q13. Segmentation is first observed in:**Ans: **(c) Annelida** (Metameric segmentation).**Q9. "All vertebrates are chordates but all chordates are not vertebrates." Justify.**

Ans: All chordates possess a notochord. In vertebrates, this embryonic notochord is replaced by a bony/cartilaginous vertebral column in adults. Protochordates retain notochord without vertebral column.

Q14. Matching Key Terms:

- Operculum → *Osteichthyes* | Parapodia → *Annelida* | Comb plates → *Ctenophora* | Radula → *Mollusca* | Choanocytes → *Porifera* | Hairs → *Mammalia*.